Clinical Practice Guideline

Venous Thrombo-Embolism (VTE) Prophylaxis

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Medical, Nursing, Pharmacy

Contents

Introduction

Venous thromboembolism (VTE) remains a major cause of morbidity and a significant cause of mortality in hospitalised patients across Australia[1]. This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to Venous Thromboembolism Prophylaxis. It ensures the delivery of a uniform, integrated multidisciplinary approach with management that is consistent with best practice in accordance with the Australian Commission on Safety and Quality in Health care (ACSQHC): Venous Thromboembolism Prevention Clinical Care Standard[2].

Risk Statement

Non-compliance with this policy will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☐
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGOPHCG Mission & Values	☐	Staff Safety	☐
Other:	☐		

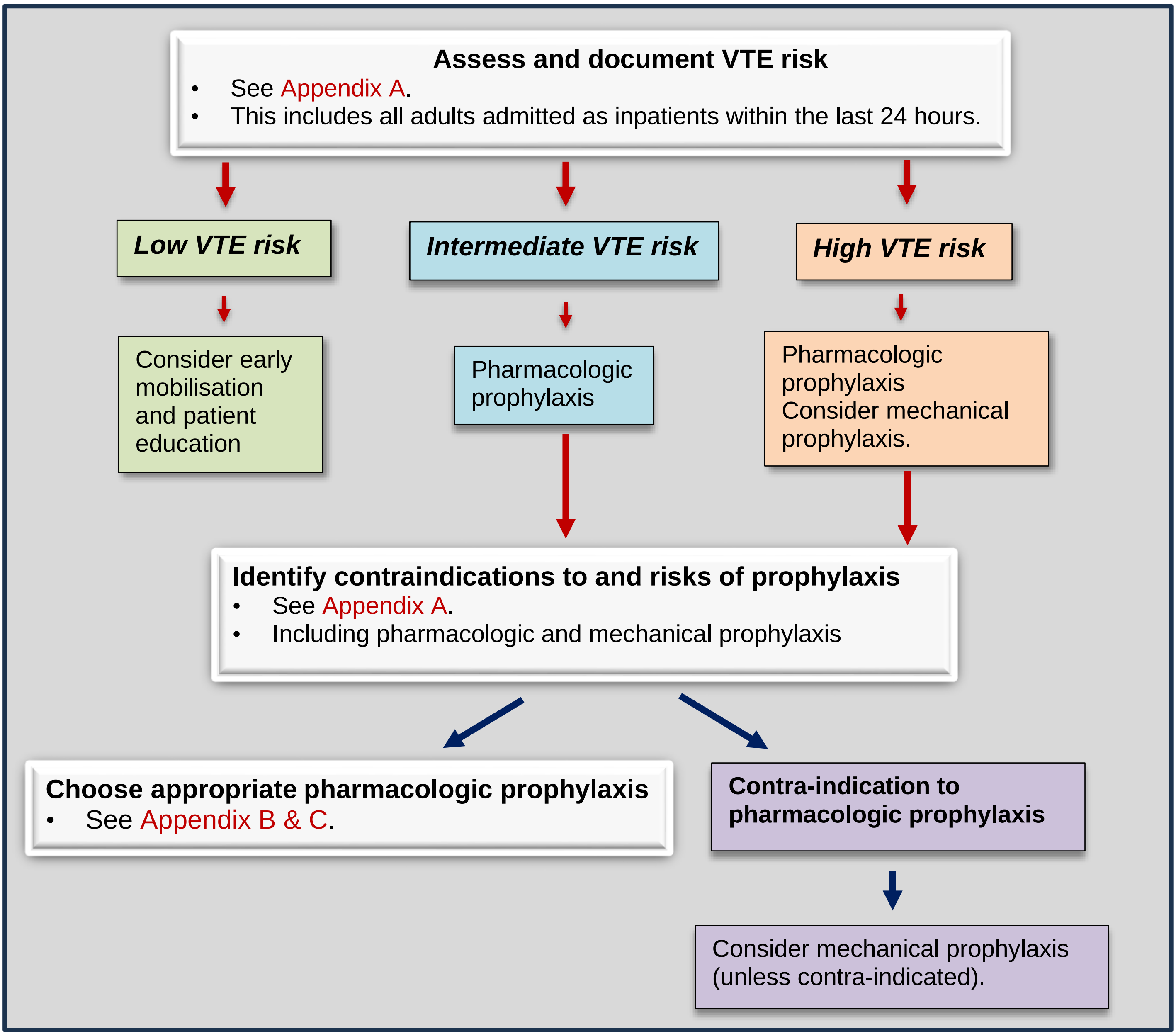
Policy Principles

- All patients admitted to Sir Charles Gairdner Osborne Park Health Care Group (SCGOPHCG) must have a Venous Thromboembolism (VTE) risk assessment documented on the Western Australia Hospital Medication Chart by the admitting Medical Officer.
- Patients will receive the most appropriate thromboprophylaxis based on an assessment of the relative risks and benefits of approved evidence-based strategies.
- The VTE risk, relative contraindications and prescribed treatment will be documented in the clinical notes.
- The ultimate responsibility of the VTE and bleeding risk assessment falls on the treating physician.

This Protocol is based on The Australian Commission of Safety and Quality clinical standards for Venous thromboembolism 2020[2]. The principles of VTE prevention are:

1. **Assess and document VTE risk on admission.**
   - Assess VTE risk on admission.
   - See Appendix A VTE risk assessment protocol
2. **Assess bleeding risk**
   - Assess risk of bleeding and contraindications to thromboprophylaxis
   - See Appendix A VTE risk assessment protocol
3. **Develop a VTE prevention plan**
   - Balance the risk of VTE against the risk of bleeding. Prophylaxis needs to be individualised
   - See Appendix B & Appendix C for prophylaxis recommendations
4. **Inform and partner with patients**
   - Inform patients of risk of VTE and side effects of prophylaxis
   - Inform patients of symptoms of VTE
5. **Document and communicate the VTE prevention plan.**
   - Document prophylaxis on WA Health Anticoagulation Chart & progress notes.
6. **Reassess risk and monitor the patient for complications.**
   - Reassess bleeding & VTE risk within 7 days, when the patient's condition changes, and on discharge.
7. **Transition from hospital and ongoing care**

- Patients considered at risk of VTE post discharge should have a documented VTE prophylaxis plan
- Educate patient how to administer the thromboprophylaxis



1. Assess and document VTE risk

To identify individual patient risk of VTE and relevant contraindications to thromboprophylaxis, all SCGH multi-day stay surgical and medical patients should be screened as soon as possible and within 24 hours of admission[2].

The following patients should have a VTE risk assessment:

- Adult patients to be discharged home from the Emergency Department who, as a result of their acute illness or injury, have significantly reduced mobility relative to normal.
- All adult patients admitted to an inpatient ward, medical or surgical, or unit including sub-acute facilities such as rehabilitation and palliative care.
- All adult patients undergoing day surgeries or procedures under general and prolonged anaesthesia with significant reduction in their mobility.

The VTE Risk Assessment undertaken by Medical Staff must be conducted using the SCGOPHCG Venous Thromboembolism Risk Assessment Protocol *(Appendix A)*.

- The assessment should be documented on the WA Health Medication Chart (WAHMC).
- Obtain the best possible medication history on admission to identify the patient's current medicines that are associated with an increased current risk of thrombosis or bleeding. This will be verified by the clinical pharmacist on the Medication History and Management Plan form.

Areas/Units with specific protocols regarding VTE risk assessment and prophylaxis that differ from the agreed Protocol are required to have these protocols approved by the SCGH Medical Executive Committee prior to their use.

2. Assess bleeding risk and contra-indications to thromboprophylaxis

Patients should be assessed for bleeding risk and contra-indications to thromboprophylaxis prior to prescription of thromboprophylaxis (pharmacologic and non-pharmacologic). See *Appendix A*.

3. Develop a VTE prevention plan

The decision to prescribe VTE prophylaxis needs to be tailored to the individual patient, after carefully balancing the risk of thrombosis against the risk of bleeding.

Treatment recommendations made in this document do not cover all clinical scenarios and still require clinical judgement.

Medications, doses and indications for pharmacological agents are outlined in *Appendix B*.

Patients must remain adequately hydrated (unless contraindicated due to their clinical condition e.g., fluid restriction). They must be encouraged to mobilise as soon as possible and to continue to mobilise post discharge[2,3].

4. Inform and partner with patients

Patients are to be informed and educated about their risk of VTE on their admission to SCGOPHCG and given the opportunity to be partners in their own care by participating in informed discussions about their VTE prevention plan.

All patients (and/or carers) should be given a SCGOPHCG 'Preventing blood clots, deep vein thrombosis and pulmonary embolism: A guide for patients and carers' pamphlet, or other locally relevant document soon after admission to a SCGOPHCG facility.

Prior to discharge the patient and / or their family and carers will be provided with education and assessment to ensure they have the capability to continue ongoing community thromboprophylaxis[3] if required.

5. Document VTE prevention plan

Once the VTE risk assessment is completed and communicated on the WAHMC:

- If pharmacological prophylaxis is required, it is to be prescribed on the WA Anticoagulation chart.
- The use of Mechanical VTE prophylaxis is to be supported by clear documentation within the patient's medical records. Compliance with mechanical prophylaxis is documented via the Comprehensive Care Plan, specialty clinical pathway, or ICU Flowchart.

6. Reassess risk and monitor the patient for VTE-related complications

During hospitalisation, a patient's bleeding and thrombotic risk should be reassessed and documented at intervals no longer than every 7 days, or sooner if the patient's clinical condition changes[2].

7. Transition from hospital and ongoing care

Patients considered at risk of VTE following discharge should have a documented VTE prophylaxis plan. Any ongoing community prophylaxis requirements are to be clearly outlined to both the patient and their general practitioner on discharge from SCGOPHCG facilities.

8. Mechanical prophylaxis

Nursing assessment

Nurses may commence graduated compression stocking (GCS) therapy in high-risk patients who do not have a contraindication. Patients will still require a documented VTE assessment on the WAHMC by a medical officer within 24 hours of inpatient admission.

Measuring and size selection

Mechanical prophylaxis products may be provided by several different suppliers.

- Follow the manufacturer's instructions for measuring and applying the requested product. Document the measurements in the patient's integrated medical record, or as per the unit's practice.
- Obtain new measurements if patients develop new swelling or oedema (e.g., post-surgery) and an appropriately sized garment will need to be refitted[3].
- GCS may be chosen based on length, open toe, closed toe or inclusion of non-slip soles.

Incorrect fitting of mechanical prophylaxis has been identified to cause patient harm[4]. If the correct size cannot be sourced, please discuss with your CNS / Manager regarding ordering the correct size. Do not apply or supply incorrectly fitting products.

Initiation

- Mechanical prophylaxis can be commenced as soon as practical after the VTE risk assessment and should be initiated prior to any surgical intervention[5,6,7].

- If Intermittent Pneumatic Compression (IPC) is to be commenced for an acute stroke it should be commenced within 3 days of the acute stroke[3].

Ongoing Care

- The chosen mechanical prophylaxis strategy should continue both day and night with minimal interruption until switched or ceased[2,3].
- Mechanical prophylaxis products should be removed to assess patient's skin integrity:
  - Once per shift in patients on continuous bed rest or with significant immobility, poor skin integrity, neuropathy or sensory loss.
  - Once a day on all other patients[3].
  - .I
  - IPC devices that connect to a stationary pump must be disconnected for mobilisation and will need them to be reapplied upon the patient's return to bed or chair[5].
- Graduated compression stockings:

  - If using heel protectors, place them outside of the GCS.
  - Patients wearing GCS without non-slip soles should not ambulate without safe footwear or other non-slip options.

- Mechanical measures may require a medical review and possible switching or early cessation if a patient develops known adverse effects to mechanical measures, e.g. altered sensation, pain or discomfort, signs of markings, blistering or discolouration or increased oedema[3,8].

Cessation

- If the patient is on IPC and has been admitted for an acute stroke, then the IPC should continue for 30 days or until the patient is mobilising independently[3].

- Medium VTE risk patients with a prior contraindication to chemical prophylaxis can be transferred to chemical prophylaxis when safe to do so. At this time these patients may have their mechanical prophylaxis ceased in the presence of documented instructions. High risk VTE patients may benefit from continuing both measures[5,7].
- For medical patients and low thrombotic risk surgical patients, in the absence of ongoing instructions mechanical prophylaxis can be stopped on patient discharge[8].

All GCS garments and IPC sleeves are single patient use only and should be discarded after patient discharge or therapy cessation. Please note that IPC products may have reusable tubing which need to be saved and cleaned between patients (seek manufacturer information to clarify).

Compliance, monitoring and evaluation

Clinical Division Co-Directors and Heads of Departments are responsible for ensuring compliance with policy.

SCGOPHCG monitors compliance with this guideline through local quality improvement activities and routine data collection on Hospital Acquired Complications (HACs).

Related Documents

- Australian Commission on Safety and Quality in Healthcare [ACSQHS]. Venous Thromboembolism Prevention Clinical Care Standard January 2020 (safetyandquality.gov.au)
- Department of Health WA:
  - High Risk Medication Policy; WA Anticoagulation Medication Chart (MP 0131/20)
  - Western Australian Medication Chart
  - Western Australian Anticoagulation Chart (SCGH MR401/805.2 / OPH MR172)
  - SCGOPHCG HG002: Footwear for Patient Safety

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Authorisation & Version Control

Sponsor:	Director of Clinical Services				
Policy Contact:	Anticoagulation Nurse Practitioner/ Haematology Thrombosis Consultant				
Version #:	2	**Last Reviewed:**	Oct 2025	**Review Due:**	14/10/2028
Reference #	CPG008	**Consumer Engagement:**	N/A	**Aboriginal Health ISD #:**	N/A
Approved by:	Director of Clinical Services			**Date:**	21/07/2025
Endorsed by:	SCGOPHCG Clinical Practice Guideline Sub Committee			**Date:**	14/10/2025
Applicable Standards	NSQHS Standards Std 1: Clinical Governance | Std 2: Partnering with Consumers | Std 4: Medication Safety | Std 5: Comprehensive Care | Std 6: Communicating for Safety | Std 8: Recognising and Responding to Acute Deterioration				
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Appendix A: VTE Risk Assessment Protocol

This protocol is based on the NSW Clinical Excellence Commission (CEC) tool[10]

Determine Risk Factors for thrombosis and allocate into a risk category

Higher VTE risk

- ☐ Total hip replacement, total knee replacement
- ☐ Hip fracture surgery
- ☐ Abdominal or pelvic surgery for cancer
- ☐ Multiple major trauma
- ☐ Acute spinal cord injury with paresis

Intermediate VTE risk

- ☐ Patients not in lower or higher VTE risk group

Lower VTE risk

- ☐ Ambulatory patient without VTE risk factors
- ☐ Non-surgical ambulatory patient with VTE risk factors and expected length of stay ≤2 days
- ☐ Minor surgery* without VTE risk factors

*same day surgery or operating time < 30 min

Consider VTE risk factors

VTE risk factors

- ☐ Age > 60 years
- ☐ Obesity (BMI > 30 kg/m^2)
- ☐ Moderate to major* surgery

  *Operating time > 45 min or involves the abdomen
- ☐ Prior history of VTE
- ☐ Known thrombophilia (including inherited disorders)
- ☐ Active malignancy or cancer treatment
- ☐ Myeloproliferative neoplasms
- ☐ Acute myocardial infarction
- ☐ Congestive cardiac failure
- ☐ Active or chronic lung disease
- ☐ Active infection
- ☐ Active rheumatic disease
- ☐ Acute inflammatory bowel disease
- ☐ Hormone replacement therapy
- ☐ Oestrogen-based contraceptives
- ☐ Nephrotic syndrome
- ☐ Dehydration
- ☐ Varicose veins/chronic venous stasis
- ☐ Significantly reduced mobility relative to normal state
- ☐ Pregnant or post-partum (refer to obstetric team)
- ☐ Sickle cell disease

Identify contraindications and patient risks related to pharmacologic prophylaxis

Absolute Contraindications

- Thrombocytopenia (platelets <50 x 10^9/L) or severe platelet dysfunction
- Coagulopathy
- Therapeutic anticoagulation
- Active haemorrhage

Relative contraindications

- Craniotomy within two weeks
- Intraocular surgery within 2 weeks
- Gastrointestinal or genitourinary haemorrhage within the last month
- Intracranial haemorrhage within last year
- Active intracranial neoplasms
- Use of antiplatelet agents
- Inherited bleeding disorders
- High Falls risk
- Severe trauma to head or spinal cord with haemorrhage
- End stage liver disease (INR > 1.5)
- Hypertensive emergency

Other conditions to consider

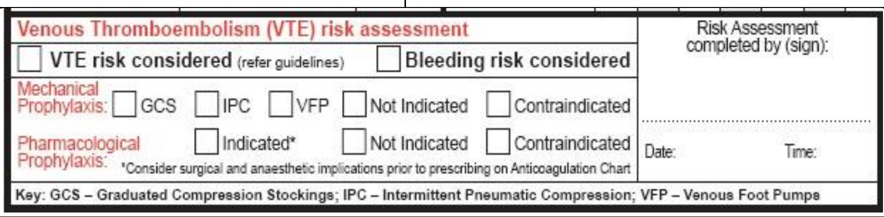
- Recent neuraxial anaesthesia or recent lumbar puncture
- Heparin sensitivity or history of heparin induced thrombocytopenia (HIT)
- Creatinine clearance < 30ml/min
- Extremes of weight (<50kg or > 120kg, BMI >35kg/m^2)

Identify contraindications to mechanical prophylaxis

- Morbid Obesity or severe lower limb deformity (where the correct sizing of stockings cannot be achieved)
- Acute stroke (Intermittent Pneumatic Compression can be used)

- Severe oedema of the legs
- Severe Falls Risk
- Recent skin graft
- Severe dermatitis, skin ulceration or infected wounds or gangrene (consider venous foot pumps)

- Severe Peripheral Artery Disease
- Peripheral Neuropathy (Intermittent Pneumatic Compression can be used)
- Allergy to product materials
- Compartment syndrome



Venous Thromboembolism (VTE) risk assessment

☐ VTE risk considered (refer guidelines) ☐ Bleeding risk considered

Risk Assessment completed by (sign):

Mechanical Prophylaxis: ☐ GCS ☐ IPC ☐ VFP ☐ Not Indicated ☐ Contraindicated

Pharmacological Prophylaxis: ☐ Indicated* ☐ Not Indicated ☐ Contraindicated

*Consider surgical and anaesthetic implications prior to prescribing on Anticoagulation Chart

Date: Time:

Key: GCS – Graduated Compression Stockings; IPC – Intermittent Pneumatic Compression; VFP – Venous Foot Pumps

Instructions: The Doctor completing must tick the check box acknowledging they have considered VTE risk and Bleeding risk before noting the indications for prophylaxis.

Appendix B: Prophylaxis Recommendations by Procedure / Condition (Non-Cancer)

Pharmacologic prophylaxis: standard recommendations			
Medication	**Dose**	**Duration**	**Comment**
Enoxaparin (LMWH)	40mg subcutaneously daily eGFR <30ml/min: 20mg subcutaneously daily	Medical: until patient is no longer at risk of VTE (i.e. until mobility returns to normal or discharge) Surgical: see below	For VTE prophylaxis at body weight <50kg or >120kg, or BMI >35kg/m^2 seek specialist advice.
Unfractionated heparin (UFH)	5000U subcutaneously 8 hourly* or 12 hourly		*8 hourly is associated with a higher bleeding risk

Mechanical prophylaxis recommendations	
Indications	In surgical patients at high risk of VTE, with pharmacologic prophylaxis In patients at moderate or high risk of VTE, in whom pharmacologic prophylaxis is contra-indicated.
Duration	Until mobility and activity returns to baseline[3].

Specific recommendations			
Indication	**Drug/dose**	**Duration**	**Comment**
Total hip replacement/hip fracture surgery	LMWH OR Rivaroxaban 10mg daily or Apixaban 2.5 mg BD	28 to 35 days	UFH not recommended
Total knee replacement	LMWH or Rivaroxaban 10mg daily or Apixaban 2.5 mg BD	10 - 14 days	
Lower leg immobilisation due to injury	LMWH	Until mobility returns to baseline	
Major general surgery	LMWH or UFH	Continue for up to 7 days or until mobility returns to baseline	
Past history of Heparin Induced Thrombocytopenia (HIT)	Fondaparinux: CrCl > 50ml/mL/min: 2.5mg daily CrCl < 50ml/min: consult haematology	Until discharge	Discuss with the haematology registrar.

Notes:

1. VTE prophylaxis dose for total body weight <50kg or > 120Kg: seek specialist advice.
2. *Caution:* Review indication/dosing when prescribing LMWH or Rivaroxaban when creatinine clearance <30ml/min.
3. If central neural blockade is used there is a risk of developing an epidural haematoma. To minimise this risk, timing of pharmacological thromboprophylaxis should be carefully planned and discussed in advance with the anaesthetist.
4. Aspirin and warfarin are not recommended as the sole agent for VTE prophylaxis[7].
5. All Patients require an assessment of their individual risk factors and contraindications to prophylaxis.

Appendix C: Prophylaxis Recommended for Cancer Patients[11,12]

	Pharmacologic prophylaxis	Mechanical prophylaxis
Surgical	• LMWH for 7-10 days for general surgical procedures including abdominal, pelvic surgery or neurosurgery for Cancer. • Consider extended thromboprophylaxis (LMWH for up to 28 days) post discharge in patients post major abdominal or pelvic surgery for cancer, if the bleeding risk is low.	GCS if pharm prophylaxis contraindicated (e.g. if the bleeding risk is high).
Medical	• LMWH from admission until discharge. • For ambulatory patients with cancer at high risk of thrombosis, consider thromboprophylaxis. • For patients with multiple myeloma, on lenalidomide, pomalidomide or thalidomide, consider using low dose aspirin or LMWH.	GCS if pharm prophylaxis contraindicated